

product (patient and user) safety risk (referred to as system safety in this handbook, see [Section 3.1.11](#)).

Evolving Risk and Opportunity Management Concepts. According to Conrow (2003), “Traditionally, risk has been defined as the likelihood of an event occurring coupled with a negative consequence of the event occurring. In other words, a risk is a potential problem—something to be avoided if possible, or its likelihood and/or consequences reduced if not.” As a corollary to risk, Conrow (2003) defines opportunity as “the potential for the realization of wanted, positive consequences of an event.” The idea of considering opportunities and positive outcomes (in addition to negative outcomes) as an integral part of a Risk Management process has gained favor with some experts and practitioners. New risk and risk management concepts intended to support this broadened scope for risk management are evolving.

The measurement of risk has two components (see [Figure 2.28](#)):

Consequence rating ↑	a	III	III	II	I	I
	b	IV	III	III	II	I
	c	V	IV	III	II	I
	d	V	V	IV	III	II
	e	V	V	IV	III	II
		1	2	3	4	5
		Likelihood rating →				

FIGURE 2.28 Level of risk depends upon both likelihood and consequence. From ISO/IEC 31010 (2019). Used with permission. All other rights reserved.

- The likelihood that an event will occur
- The undesirable consequence of the event if it does occur

The generic consequence/likelihood matrix in [Figure 2.28](#) is a way to display risks according to their consequence (illustrated with the generic a-e, high-to-low, consequence rating scale), and their likelihood (illustrated with the generic 1-5, low-to-high, likelihood rating scale), and to combine these characteristics to display a rating for risk level (illustrated with the generic Roman numeral I-V, high-to-low, risk significance scale). The combination of low likelihood and low undesirable consequences gives low risk, while high risk is produced by high likelihood and highly undesirable consequences. Risk prioritization and decision rules (such as the level of management attention or the urgency of response) can be linked to the matrix cells. Note that this generic matrix is conceptual and cannot be applied without careful customization to address the specific project. Detailed guidance and examples for designing rating scales and matrices suitable for use on specific projects, products and systems are provided in IEC 31010 (2019).

A positive consequence scale may be used in the matrix shown in [Figure 2.28](#), thereby changing the outcome adjective from undesirable to desirable, and the cells in the matrix from risks to opportunities. Note that the foundational concept and structure of the matrix diagram remains the same.

SE and project management are all about pursuing an opportunity to solve a problem or fulfill a need. Opportunities enable creativity in resolving concepts, architectures, designs, and strategic and tactical approaches, as well as the many administrative issues within the project. It is the selection and pursuit of these strategic and tactical opportunities that determine just how successful the project and system will be. Of course, opportunities usually carry risks, and each opportunity will have its own set of risks that must be intelligently judged and properly managed to achieve the full value (Forsberg, et al., 2005). These are the risks that must be managed to enhance the opportunity value and the overall value of the project (see [Figure 2.29](#)). Opportunity management and risk management are therefore essential to—and performed concurrently with—the planning process but

require the application of separate and unique techniques that justify this distinct technical management element.

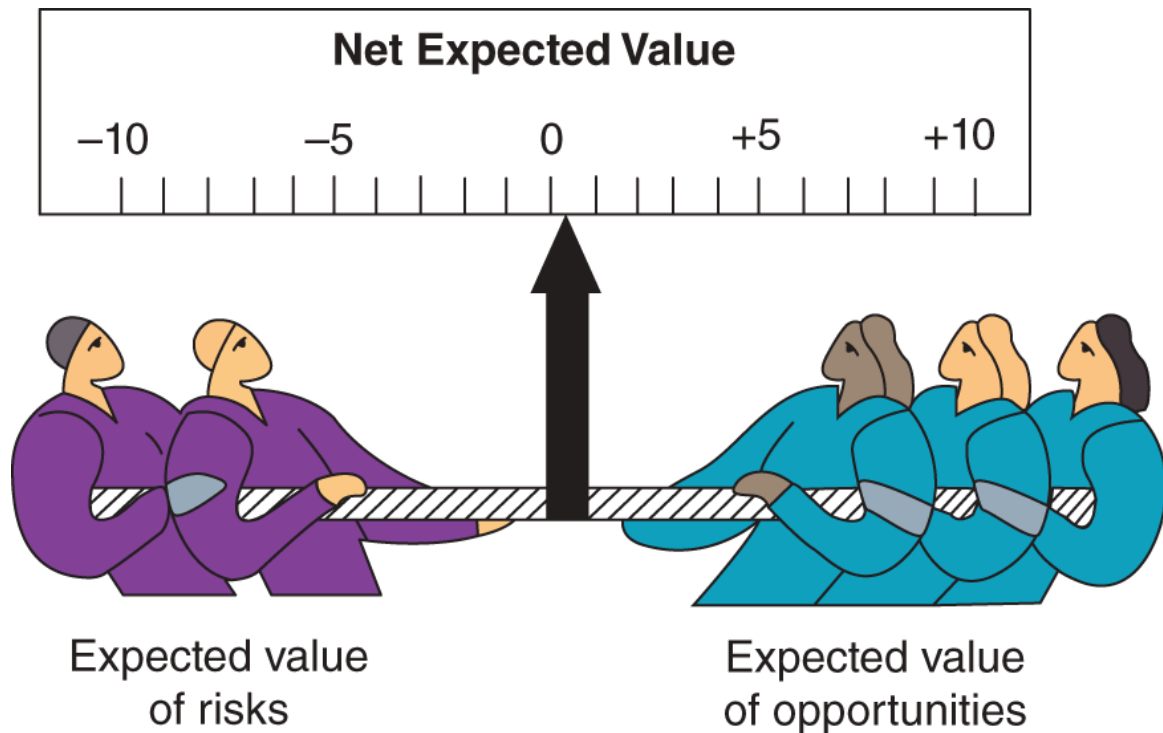


FIGURE 2.29 Intelligent management of risks and opportunities. From Forsberg, et al. (2005) with permission from John Wiley & Sons. All other rights reserved.

Balancing Project, Risk, and Opportunity Management for SE. No realistic project can be planned without risk. The challenge is to define the system and the project that best meet overall requirements, allow for risk, and achieve the highest chances of project success. [Figure 2.30](#) illustrates the major interactions between the four risk categories: technical, cost, schedule, and programmatic. The arrow labels indicate typical risk relationships, others are possible.

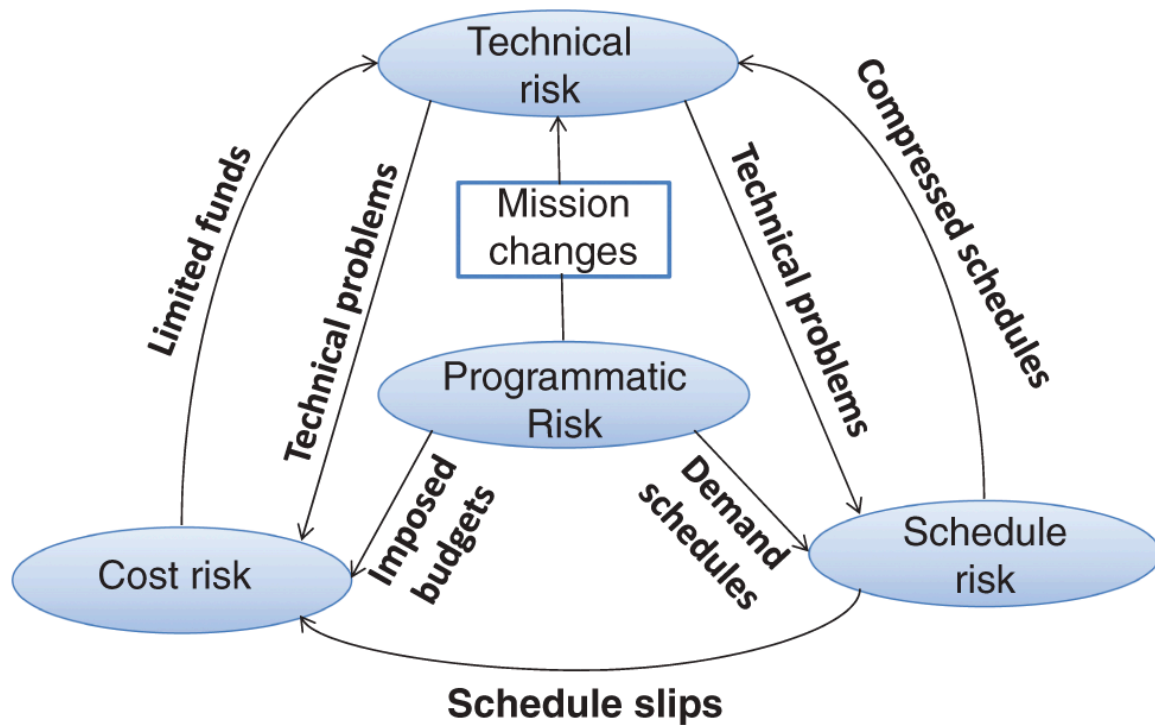


FIGURE 2.30 Typical relationship among the risk categories. INCOSE SEH original figure from INCOSE SEH v1 Figure 4.5–7. Usage per the INCOSE Notices page. All other rights reserved.

The Risk Management process is used to understand the potential cost, schedule, and performance (i.e., technical) risks associated with a system, and then take a (proactive) structured approach to anticipate negative outcomes and respond to them before they occur. With respect to opportunities, this process is used to understand the potential cost, schedule, and performance (i.e., technical) improvement opportunities associated with a system, and then take a (proactive) structured approach to defining potential positive outcomes and responding to them by adopting the best candidate improvements before the “window of opportunity” is missed. Care is taken to consider new and increased risk created as a result of pursuing a new opportunity. This practice can help identify unintended negative consequences that might be introduced by the proposed change.

Integrating Risk Management. Per ISO/IEC 31000, “integrating risk management with all organizational processes improves the performance of risk management while gaining efficiencies.” Section 7 of ISO/IEC/IEEE 16085 “Risk management in life cycle processes” provides a methodical approach for the integration of risk management and “risk-based thinking” into all SE life cycle processes. Organizations typically manage risks and

opportunities of many types, across and throughout the organization. Risks and opportunities associated with system development should be managed in a manner consistent with the organization's overall risk and opportunity management strategies.

Risk Management and the System Life Cycle. Once the scope and context of a system have been established from a hierarchical standpoint, it is possible to define and model the system (and its associated risks) in relation to its life cycle, i.e., the differences in the risks in different life cycle stages. For example, risks in the concept stage are quite different than the risks in the retirement stage. It is often necessary to consider risks in other stages while performing activities in the current stage.

Risk Assessment Techniques. ISO/IEC 31010, *Risk management—Risk assessment techniques*, provides detailed descriptions and application guidance for over 30 assessment techniques ranging from brainstorming and checklists to Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis (FTA), Monte Carlo simulation, and Bayesian statistics and Bayes nets. Although a comparable set of (published) techniques for opportunity management is not available, it is notable that ISO/IEC 31010 is not without mention of opportunity, and contains the Strength, Weakness, Opportunity, and Threat (SWOT) Analysis technique. In addition, many of the techniques in ISO/IEC 31010 can be used to assess positive outcomes as well as negative outcomes. For example, FTA can be used to perform Success Tree Analysis, and techniques such as brainstorming, checklists, Monte Carlo simulation, and Bayesian statistical analysis are broadly used for most any purpose, including the assessment of opportunities. A variant of SWOT analysis that is not mentioned in ISO/IEC 31010 is Threats, Opportunities, Weaknesses, and Strengths (TOWS), which puts the emphasis on the external environment (threats and opportunities) rather than on the internal environment (strengths and weaknesses).

Risk Treatment Approaches. Risk treatment approaches (also referred to as risk handling approaches) are often established for the moderate- and high-

risk items identified in the risk analysis effort. These activities are formalized in the RMP. There are four basic approaches to treat risks:

1. *Acceptance*: Accept the risk and do no more.
2. *Avoidance*: Avoid the risk through change of requirements or redesign.
3. *Control (or Mitigation)*: Taking actions to reduce the risk by expending budget and/or other resources to reduce likelihood and/or consequence over time.
4. *Transference*: Transfer the risk by agreement with another party that it is in their scope to treat. Look for a partner that has experience in the dedicated risk area.

The following are some of the steps that can be taken to avoid or control unnecessary risks:

- *Requirements scrubbing*—Requirements that significantly complicate the system can be scrutinized to ensure that they deliver value equivalent to their investment. Find alternative solutions that deliver the same or comparable capability.
- *Selection of most promising options*—In most situations, several options are available. A trade study can include project risk as a criterion when selecting the most promising alternative.
- *Staffing and team building*—Projects accomplish work through people. Attention to training, teamwork, and employee morale can help avoid risks introduced by human errors.

For high-risk technical tasks, risk avoidance is insufficient and can be supplemented by the following approaches:

- Early procurement
- Initiation of parallel developments
- Implementation of extensive analysis and testing
- Contingency planning

For each risk that is determined credible after analysis, a Risk Treatment Plan should be created that identifies the risk treatment strategy, the trigger

points for action, and any other information to ensuring the treatment is effectively executed. The Risk Treatment Plan can be part of the risk record on the risk profile. For risks that have significant consequences, a contingency plan should be created in case the risk treatment is not successful. It should include the triggers for enacting a contingency plan.

Risk Monitoring. Project management uses measures to simplify and illuminate the Risk Management process (see [Figure 2.34](#)). Measures can help identify new risks, as well as provide insight into the effectiveness of the risk treatments.

Each risk category has certain indicators that may be used to monitor project status for signs of risk. Tracking the progress of key system technical parameters can be used as an indicator of technical risk. The typical format in tracking technical performance is a graph of a planned value of a key parameter plotted against calendar time. A second contour showing the actual value achieved is included in the same graph for comparative purposes. Cost and schedule risk are monitored using the products of the cost/schedule control system or some equivalent technique. Normally, cost and schedule variances are used along with a comparison of tasks planned to tasks accomplished.

2.3.4.5 Configuration Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.3.5.1] The purpose of the Configuration Management (CM) process is to manage system and system element configurations over their life cycle.

CM establishes and maintains consistency, integrity, traceability, and control of a product's configuration. CM provides enduring truth, trust and traceability across the full life cycle of the product. Appropriate CM across

the enterprise and its supply chain, provides efficient, effective, lean, resilient, financially responsible, mature, need realization and sustainment through quantified knowledge and insight. Inadequate CM increases risk to the product—see the example in [Section 6.1](#) about the case of the Therac-25.

Description: Configuration Management (CM) is a Technical Management Process applying appropriate processes, resources, and controls, to establish and maintain consistency between product configuration information, and the product (SAE-EIA 649C).

Evolving system requirements, technology and the operating environment are a reality that must be addressed over the life of a system development effort and throughout the utilization and support stages. Furthermore, CM extended to the enterprise level supports the internal goals needed to achieve an efficient, effective, lean, and resilient enterprise.

Configuration management helps ensure:

- that product functional, performance, and physical characteristics are properly identified, documented, controlled, validated, and verified to establish product integrity;
- that changes to these product characteristics are properly identified, reviewed, approved, documented, and implemented;
- that the products produced against a given set of data are known, verified and validated.

Inputs/Outputs The functional model for the Configuration Management process is listed in [Figure 2.31](#). Descriptions of each input and output are provided in Appendix E.

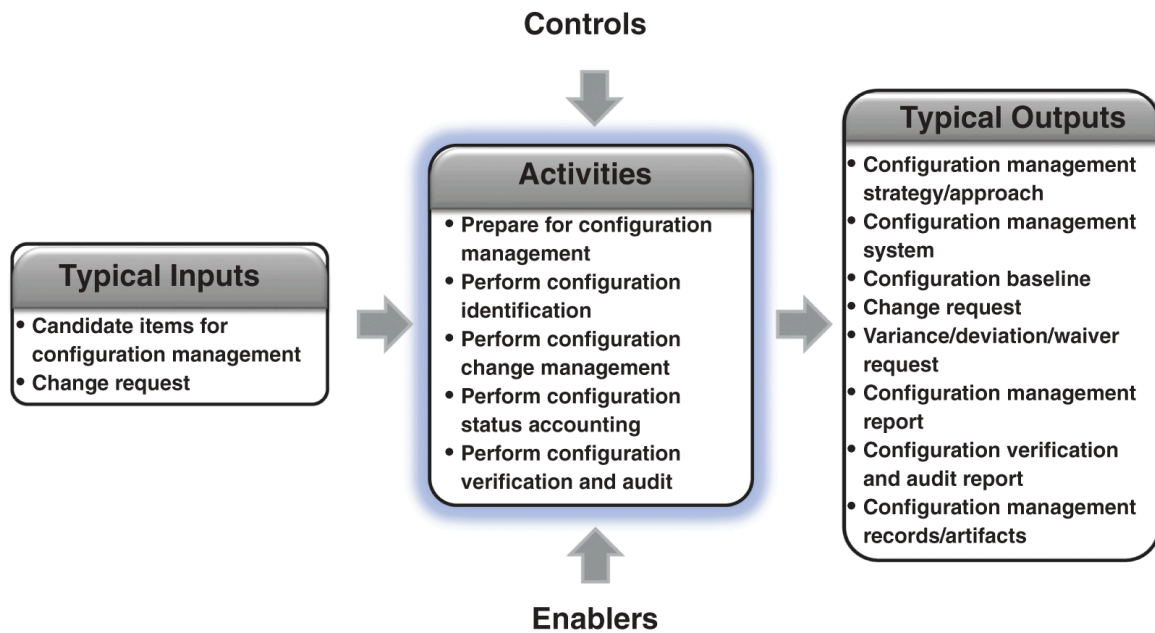


FIGURE 2.31 IPO diagram for Configuration Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Configuration Management process includes the following activities:

- *Prepare for configuration management.*
 - Similar to other SE processes, configuration management needs to be planned as early as possible in the product life cycle. The result of CM planning could be a standalone configuration management strategy, could be incorporated in the SEMP, or could be part of the digital implementation of these principles throughout the development platform (e.g., in software development where we have an integrated platform).
 - Planning and managing configuration management is accomplished in conjunction with and integrated through other SE activities and should include the following:
 1. Identify the context and environment of the system that we want to apply this to
 2. Applying adequate configuration management resources and assigning responsibility
 3. Establishing performance and status measurements
 4. Establish, implement and maintain procedures
 5. Configuration management training

6. Assessing compliance and effectiveness
 7. Supplier configuration management
 8. Product configuration information processes establishment (inc. collection and processing, controlling status, providing interoperability and exchange, long-term preservation)
 9. Planning for configuration identification, configuration change management, configuration status accounting, configuration verification, and audit.
- *Perform configuration identification.*
 - – Identify the items or elements of a system and associated data which should be under configuration management.
 - – Establish unique identifiers for the items and data under configuration management.
 - – Structure the items and information under configuration management.
 - – Validate and release items and information under configuration management.
 - – Establish and identify baselines at appropriate points throughout the life cycle. Baselines may coincide with a project milestone or decision gate as shown in [Figure 2.4](#).
 - – Manage interfaces and the constraints they impose.
 - – Identify where distinct configuration control is required (governance) and designate Configuration Items (CIs).
 - *Perform configuration change management.*
 - – Manage changes and variances (i.e., non-conformances) throughout the system life cycle to ensure the integrity of the product/system.
 - – This includes the identification, recording, review (incl. impact analysis), approve/disapprove, tracking, processing, implementing and closing of requests for change/variances, including relevant supporting documentation, whatever its origin.
 - – Disposition of changes/variances are often performed by change boards (the names used in various industries like Configuration Control Board [CCB], Configuration Review Boards [CRB], etc.).
 - – An important practice for the change/variance process is to track and manage implementation activities and close the loop ensuring

that both the product and its associated information have been evolved to the current approved configuration.

- *Perform configuration status accounting.*
 - – Communicate and maintain the status of controlled events, items, and data, as well as performance of CM processes across the life cycle of the product/system to the appropriate stakeholders.
 - – Measures and means of measuring performance are established by the Project Assessment and Control process and the configuration status accounting supports these metrics and performance assessments (e.g., Performing reconciliation of the As-Designed data with the As-Built data).
- *Perform configuration verification and audit.*
 - – Perform verification of CM processes, in conjunction with the Verification and Quality Assurance processes. Verification includes: review of CM processes; verifying to ascertain that the system has achieved specified requirements and the design of the system is accurately and completely documented in configuration information; verify physical, functional, and interface requirements defined in the approved product definition information, are achieved by the product; verify approved changes to its configuration. Auditing supports the verification process by validating traceability and status between the product to its design, product design to its requirements, and the implementation of changes. Auditing at events like a functional configuration audit (FCA) or a physical configuration audit (PCA), is often accomplished at the end of the development effort and/or testing.
 - – Furthermore, the acquirer may have the requirement or wish to perform surveillance and, where necessary, audits to ensure the correct application of CM processes in their supply chain.

Common approaches and tips:

- Begin the Configuration Management process at the beginning of the system life cycle and continue through until retirement of the system. Tailoring of the configuration management approach is key for its

successful applications across various domains; this includes an appropriate understanding of the information and processes that need to be in place to fulfill all CM requirements.

Elaboration

Additional guidance regarding configuration management can be found in the current versions of SAE-EIA 649C, ISO 10007, and IEEE 828. Application domain-specific practices, such as SAE ARP 4754A, GEIA HB 649, MIL HDBK 61 B, NIST 800–53, NIST 800–128 provide additional application details.

Configuration management must account for horizontal and vertical integration (see [Section 2.3.5.8](#)), in addition to other factors that can affect the system definition over time. Change is a fundamental characteristic of every large-scale system during its life cycle; baselines are set, design fidelity and completeness are improved, and problems are resolved as analyses are performed, impacts are assessed, and trade studies result in decisions that change the system definition. This constancy of change as the design matures makes it imperative to understand the impact of change across all interacting elements and to ensure the complete incorporation of change decisions. Consequently, configuration management, including change management, coordinates maturation of the system.

In Model-Based Systems Engineering (MBSE), CM is required to assure and ensure that the product/system and its product configuration information (i.e., the configuration) are appropriately captured, organized, managed, and communicated for the benefit of the model's stakeholders and participants (see [Section 4.2.1](#)).

The corresponding testing and deployment provisions need to be considered in terms of checks against validation rules, interface compatibility, flow time alignment, technical performance measure evaluation, physical clashing, and other domain-specific characteristics.

Moreover, although cyber security is traditionally thought of as a software engineering problem, it needs to be taken into account in a wider system's

engineering thought process. Hardware components on which the software is deployed as well as system interfaces can be just as susceptible to cyber-attacks as software itself. That is why proper configuration management needs to also include continuous auditing of potential cyber vulnerabilities. CM processes that originated in agile software engineering (SWE), are now widely used in other Engineering disciplines, including MBSE where the most challenging aspect is the constant need to maintain the relationships between the appropriate configurations of each domain while ensuring accountability and consistency. Several well-established CM practices in agile SWE help with addressing those pain points:

- revisions are managed as a stream of commits;
- baselines are established by tagging specific commits;
- concurrent changes are managed through branching and merging;
- testing, evaluation and/or deployment are automated through a Continuous Integration and Continuous Delivery (CI/CD) process
- security is ensured through the DevSecOps life cycle by integrating security tools into DevOps (see [Figure 2.8](#))

The digital thread establishes communication paths between the individually configured domains. It is also responsible for correctly tying together the appropriate configurations in each domain and to form a consistent configuration for a specific system/product and their elements. More details on traceability can be found in [Section 3.2.3](#).

2.3.4.6 Information Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.3.6.1] The purpose of the Information Management process is to generate, obtain, confirm, transform, retain, retrieve, disseminate, and dispose of information to designated stakeholders.

Information management plans, executes, and controls the provision of information to designated stakeholders that is unambiguous, complete, verifiable, consistent, traceable, and presentable. Information includes technical, project, organizational, integration, contractual, agreement, and user information. Information is often derived from data artifacts of the organization, system, process, or project.

Information management needs to provide relevant, timely, complete, valid, and, if required, protected information to designated parties during and, as appropriate, after the product/system life cycle. It manages all defined information, including technical, project, organizational, integration, contractual, agreement, and user information.

Information management ensures that data is properly defined, stored, structured, maintained, secured, exchanged and accessible to those who need it, thereby establishing/maintaining integrity of relevant system life cycle artifacts.

Description Information exists in many forms, and different types of information have different values within an organization. Information assets, whether tangible or intangible, have become so widespread in contemporary organizations that they are indispensable. Information Security has become a fundamental requirement for every industry to work within digital environments with confidence. The following are important terms in information management:

- Information is what an organization has compiled or its employees know. It can be stored and communicated, and it may include classified or unclassified, export restrictive, proprietary, and/or protected (e.g., by copyright, trademark, or patent) and unprotected (e.g., business intelligence) intellectual property. Specific domain classification may apply as well (e.g., further classifications, like Controlled Unclassified Information (CUI) protections in the US defense domain).
- Information assets are intangible information and any tangible form of its representation, including drawings, models of all flavors (systems,

software, design, simulation, manufacturing, etc.), specifications, memos, email, computer files, and databases.

- Information security generally refers to the protection, confidentiality, integrity, and availability of the information assets (ISO 17799, 2005).
- Information security management includes the controls used to achieve information security and is accomplished by implementing a suitable set of formalized controls, which could be policies, practices, procedures, organizational structures, and software.
- Information Security Management System is the life cycle approach to implementing, maintaining, and improving the interrelated set of policies, controls, and procedures that ensure the security of an organization's information assets in a manner appropriate for its strategic objectives.

Information management must be associated very closely with configuration management to ensure the integrity, initial release and change control of the information and data. Information management provides the basis for the management of and access to information throughout the system life cycle from ideation through disposal. Designated information may include organizational, project, integration, contractual, agreement, technical, and user information. The mechanisms for maintaining historical knowledge in the prior processes—decision making, risk, and configuration management—are under the responsibility of configuration management working in concert with information management. [Figure 2.32](#) is the IPO diagram for the Information Management process.

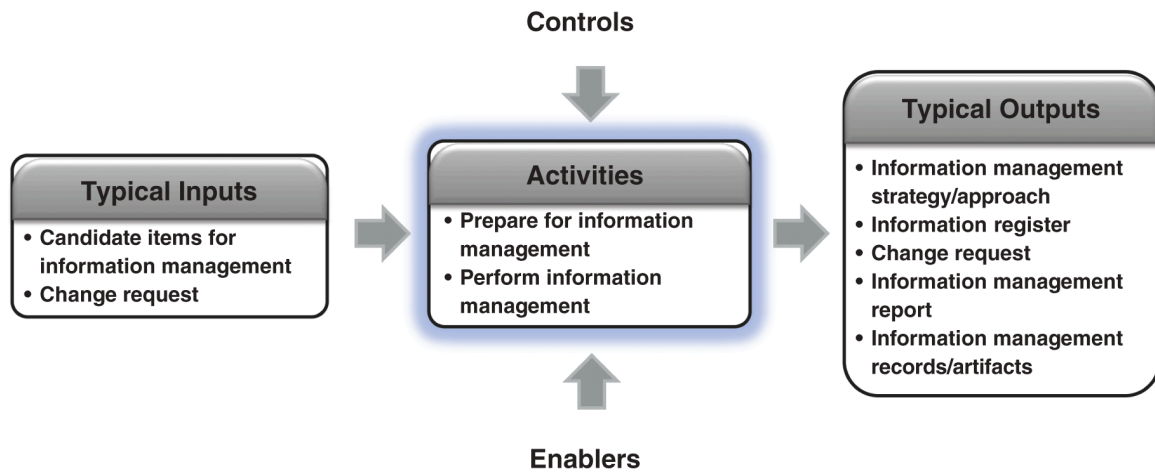


FIGURE 2.32 IPO diagram for Information Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Knowledge management is also closely linked to information management. One of the motivations for putting knowledge management in place is for “Information sharing across the organization” thus information management is key for knowledge management. Whereas “Share knowledge and skills throughout the organization” as one of the process-activities for knowledge management draws an identifiable relationship between the information management and Knowledge Management processes via the feedback loop created with inputs and outputs between the processes if mapped out.

Inputs/Outputs Inputs and outputs for the Information Management process are listed in [Figure 2.32](#). Descriptions of each input and output are provided in Appendix E.

Process Activities The Information Management process includes the following activities:

- *Prepare for Information Management.*
 - – Support establishing and maintaining a system data dictionary—see project planning outputs.
 - – Define system-relevant information, revisioning scheme, storage requirements, access privileges, and the duration of maintenance.
 - – Define formats and media for capture, retention, transmission, and retrieval of information.

- – Identify valid sources of information (e.g., business processes) and designate authorities and responsibilities regarding the origination, generation, capture, release, archival, and disposal of information in accordance with the Configuration Management process.
- *Perform Information Management.*
 - – Periodically obtain or transform artifacts of information. This is not necessarily specific to waterfall approaches. “Periodically” can also mean at each increment or each iteration/sprint.
 - – Maintain information according to integrity, security, and privacy requirements.
 - – Retrieve and distribute information in an appropriate form to designated parties, as required by agreed schedules, definitions, or defined circumstances.
 - – Archive designated information for compliance with legal, audit, knowledge retention, and project closure requirements.
 - – Dispose of unwanted, invalid, or unverifiable information according to organizational policy, security, privacy, and legal requirements applicable to the data.

Common approaches and tips:

- Identify information-rich artifacts and store them for later use even if the information is informal, such as a design engineer’s notebook (in any media or format).
- Identify the information set at the start of a project if you are going to follow a digital engineering approach.
- In the Project Planning process (see [Section 2.3.4.1](#)), an information management plan is tailored to satisfy the individual project procedures for information management. An information management plan identifies the system-relevant information to be collected, retained, controlled, secured, and disseminated, with a schedule for disposal.

Elaboration

The initial planning efforts for information management are defined in the information management plan (and should align with the Configuration Management Plan), which establishes the scope of information that is maintained; identifies the resources and personnel skill level required against the defined tasks to be performed; defines the rights, obligations, and commitments of parties for generation, management, and access; and identifies information management tools and processes, as well as methodologies, standards, and procedures that will be used on the project and managed by appropriate configuration management.

Effective information management provides readily accessible information and management means to authorized project and organization personnel. Database management, security, and revision of data, sharing data across multiple platforms and organizations are facilitated by information management. With all emphasis on knowledge management, organizational learning, and information as competitive advantage, these activities are gaining increased attention.

2.3.4.7 Measurement Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288 (and ISO/IEC/IEEE 15939),

[6.3.7.1] The purpose of the Measurement process is to collect, analyze, and report objective data and information to support effective management and address information needs about the products, services, and processes.

Description The Measurement process defines the types of information needed to support project and technical management decisions and implement actions to manage and improve performance. The key SE measurement objective is to assess the SE processes and work products with respect to project and organization needs, including timeliness, meeting performance requirements and quality characteristics, product

conformance to standards, effective use of resources, and continuous process improvement in reducing cost and cycle time.

The *Practical Software and Systems Measurement (PSM) Guide* (2003), [Section 1.1](#), states:

Measurement provides objective information to help the project manager.

Specific measures are based on information needs and how that information will be used to make decisions and take action. Measurement thus exists as part of an integrated set of management processes and includes not just the project manager, but also SE practitioners, analysts, quality management/assurance, and nearly all other technical and management functions/roles. The decisions to be made drive the information needs and the information needs drive the data to be collected, analyzed, and reported. As a result, numerous benefits are realized from effective measurement (see [Table 2.5](#)).

[TABLE 2.5](#) Measurement benefits

Benefit to Project Manager/Technical Lead

Communicate effectively throughout the project organization
Identify and correct problems early
Support making key trade-offs
Track specific project objectives
Defend and justify decisions
Enable continuous process improvement

From PSM (2003). Used with permission. All other rights reserved.

Successful measurement communicates meaningful information to the decision makers. The presentation of the information must be relevant and

unambiguous to those using it, ensuring the intended interpretation.

Inputs/Outputs Typical inputs and outputs for the Measurement process are listed in [Figure 2.33](#). Descriptions of each input and output are provided in Appendix E.

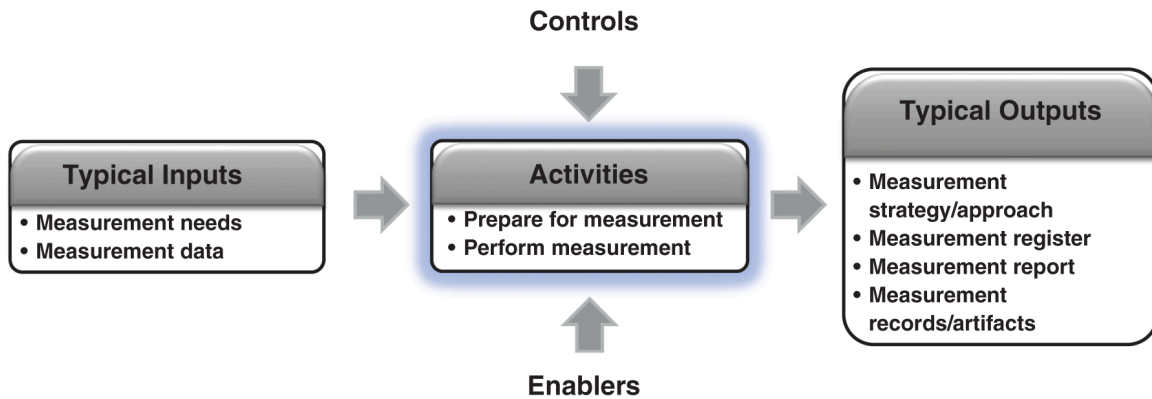


FIGURE 2.33 IPO diagram for Measurement process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Measurement process includes the following activities:

- *Prepare for measurement.*
 - – Identify the measurement stakeholders and their measurement information needs and develop a strategy to meet them.
 - – Identify and select relevant prioritized measures that aid with the management and technical performance of the project.
 - – Define the base measures, derived measures, indicators, data collection, measurement frequency, measurement repository, reporting method and frequency, trigger points or thresholds, and review authority.
- *Perform measurement.*
 - – Gather, process, store, verify, and analyze the data to obtain measurement results (information products).
 - – Record and review the measurement information products with the measurement stakeholders and recommend action, as warranted by the results.

Common approaches and tips:

- Measurement for measurement sake is a waste of time and effort. Collecting data without an information need and an intended use is not effective use of limited resources.
- Each measure should be regularly reviewed by the measurement stakeholders. The frequency of review is determined by a number of factors, including frequency of data availability/change, level of risk, maturity of the organization, and cycle times.
- Some agreements identify measures of effectiveness (MOEs) that must be met. The derived measures of performance (MOPs) and Technical Performance Measures (TPMs) that provide the necessary insight into meeting the MOEs are default measures to be included within the measurement plan. Other measures to consider should provide insight into technical and programmatic execution of the project (Roedler and Jones, 2005).
- The best measures are repeatable, can be implemented with automated data collection or require minimal effort for data collection, are straightforward to understand, and are presented in a consistent format on a regular basis (with trend data and, where applicable, projections).
- Many methods are available to present the data to the measurement stakeholders. Line graphs, control charts, and Red/Yellow/Green “traffic lights” are some of the more frequently used. Tools are available to help with measurement.
- If a need for corrective action is perceived, further investigation into the measures may be necessary to identify the root cause of the issue to ensure that corrective actions address the cause instead of a symptom.
- Measurement by itself does not control or improve process performance, project success, or product quality. Measurement results must be provided to decision makers in a manner that provides the needed insight for the right decisions to be made. Action must be taken, to realize any benefit.

Elaboration

Measurement Concepts. Measurement concepts have been expanded upon in the previous works shown in [Table 2.6](#) that the SE measurement practitioner should reference for further insights.

TABLE 2.6 Measurement references for specific measurement focuses

Reference Focus	Reference
General Reference	<i>Systems and Software Engineering -Measurement Process</i> (ISO/IEC/IEEE 15939, 2017) <i>Guide to the Systems Engineering Body of Knowledge</i> (SEBoK), Part 3: SE and Management/Systems Engineering Management/Measurement (SEBoK, 2023) <i>Practical Software and Systems Measurement (PSM) Guide V4.0c</i> , (PSM, 2003) <i>Capability Maturity Model Integration (CMMI®) for Development V2.0</i> , Measurement and Quantitative Management Process Areas (CMMI, 2018)
Guidance for New Practitioners	<i>INCOSE Systems Engineering Measurement Primer</i> , Version 2.0 (INCOSE Measurement Primer, 2010)
Technical Measurement / Performance	<i>Technical Measurement Guide</i> (Roedler and Jones, 2005)
System Development	<i>System Development Performance Measurement Report</i> (NDIA, et al., 2011)
Project Management	<i>Project Manager's Guide to Systems Engineering Measurement for Project Success</i> (INCOSE PMGtSEMfPS, 2015)

Reference Focus	Reference
Continuous Iterative Development	<i>Continuous Iterative Development Measurement Framework</i> (PSM, et al., 2021)
Digital Engineering	<i>Practical Software and Systems Measurement (PSM) Digital Engineering Measurement Framework</i> (INCOSE, et al., 2022)
Leading Indicators	<i>Systems Engineering Leading Indicators Guide, Version 2.0</i> (Roedler, et al., 2010)

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Measurement Approach. As discussed in the INCOSE Measurement Primer (2010), measurement may be thought of as a feedback control system. Value is obtained from measurement when the data analysis provides insight for assessment or action by decision makers (e.g., action is taken due to a variance from a target value or the need to improve current performance to a more desirable level). Comparing the target value and the allowable difference between the target and actual values enables decisions based upon evaluation of risk to the project or product performance meeting their required goals.

Relationship of Measurement to Risk Management and Decision Management. The measures for a project are driven by the information needs of the project and its decision makers. One source of the information needs are the objectives of the project, which can be related to resources, technical performance of the system, product or process quality, or other aspects of the project that are considered essential. Another key source of information needs are the key risks of the project. As shown in [Figure 2.34](#), the Risk Management process identifies risks that need to be monitored,

thus creating information needs that drive new measures. The Measurement process helps characterize and quantify the risks. In turn, the results of the measurement analysis may uncover new risks that need to be considered by the Risk Management process. The results of both risk management and measurement provide essential insight to decision makers that is essential to the Decision Management process. Measurement also provides insight to all other processes, especially Project Planning, Project Assessment and Control, Quality Assurance, Life Cycle Model Management, and the Technical Processes.

Digital Engineering (DE) Measurement. DE has three interrelated concerns: the transformation of engineering activities to a fully digital infrastructure, artifacts, and processes; the use of authoritative sources of truth (ASOTs) to improve the efficiency and productivity of engineering practice; and the use of MBSE practice to fully integrate system data and models with engineering, project management, and other domains and disciplines. Measurement in DE focuses on the implementation of DE transformations on projects and in enterprises, including the realization of measurable benefits in performance, effectiveness, and product quality relative to traditional engineering methods. DE measures can also serve as useful leading indicators for other product related measures. For more information, see INCOSE, et al. (2022) and [Section 5.4](#).

Continuous Iterative Development. As organizations and projects move toward incremental and evolutionary approaches for acquisition and life cycle models, measurement is key to understanding progress and quality (see [Section 2.2](#)). Measures are needed to address team, product, and enterprise perspectives. Measures are needed that balance both speed and quality that delivers a best value solution based on project objectives. For more information, see PSM, et al. (2021).

Leading Indicators. Per Roedler, et al. (2010), a leading indicator is a measure for evaluating the effectiveness of how a specific activity is applied on a project in a manner that provides information about impacts

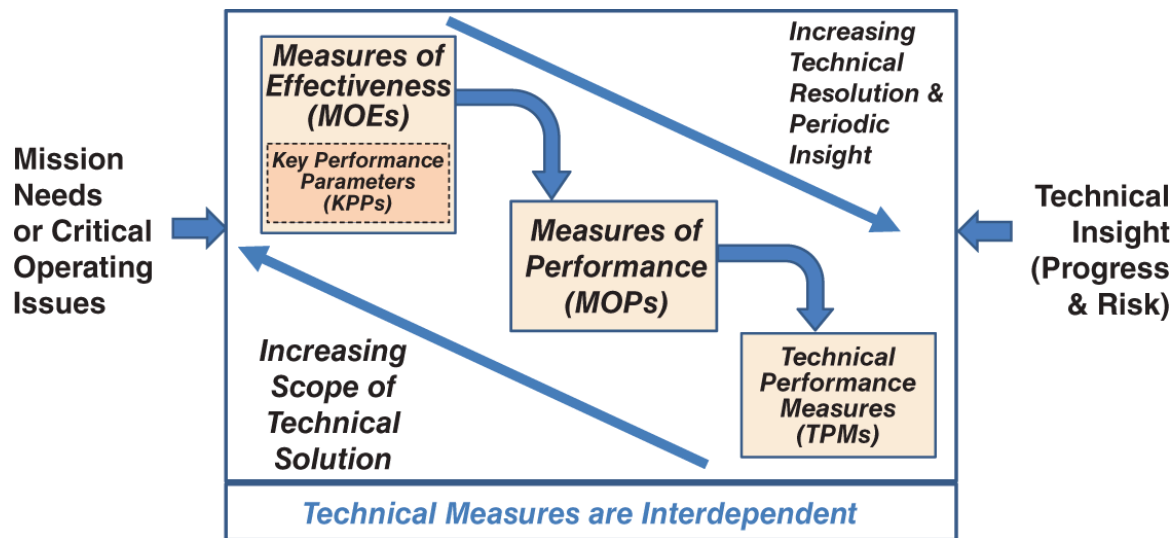
that are likely to affect the system performance or SE effectiveness objectives.

A leading indicator may be an individual measure, or collection of measures, that is predictive of future system performance before the performance is realized. Leading indicators aid leadership in delivering value to stakeholders and end users while assisting in taking interventions and actions to avoid rework and wasted effort.

Rather than provide status and historical information, leading indicators use trend information to facilitate predictive analysis (forward looking). By analyzing the trends, quantitative relationships of key factors can be developed with known correlations and predictions can be forecast on the outcomes of certain activities. Trends are analyzed for insight into both the entity being measured and potential impacts to other entities. This enables proactive decisions and actions (preventive and corrective).

For a more detailed treatment of this topic, including measurement examples, refer to Roedler, et al. (2010). In addition, NDIA, et al. (2011) provides specific leading indicators developed from the previously referenced guide for the defense and aerospace domains. However, most of the indicators have a broader application.

Product-Oriented Measures. As shown in Roedler and Jones (2005), product measures can be thought of as an interdependent hierarchy (see [Figure 2.35](#)).



Measures of Effectiveness (MOEs), which are stated from the acquirer (customer/user) viewpoint, are the acquirer's key indicators of achieving the mission needs for performance, suitability, and affordability across the life cycle. Although they are independent of any particular solution, MOEs are the overall operational success criteria (mission performance, safety, operability, operational availability, etc.) to be used by the acquirer for the delivered system, services, and/or processes.

Measures of Performance (MOPs) measure attributes considered as important to ensure that the system has the capability to achieve operational objectives. MOPs are used to assess whether the system meets design or performance requirements that are necessary to satisfy the MOEs. MOPs should be derived from or provide insight for MOEs or other user needs.

Technical Performance Measures (TPMs) are used to assess design progress, show compliance to performance requirements, and track technical risks. They provide visibility into the status of important project technical parameters to enable effective management, thus enhancing the likelihood of achieving the technical objectives of the project. TPMs are derived from, or provide insight for, the MOPs and focus on the critical technical parameters of specific architectural elements of the system as it is designed and implemented. Selection of TPMs should be limited to critical technical thresholds or parameters that, if not met, put the project at cost, schedule, or performance risk. The TPMs are not a full listing of the requirements of the system or system element. The SEMP should define the approach to TPMs (Roedler and Jones, 2005).

[Figure 2.36](#) illustrates a sample TPM. Values are established to provide limits that give early indications if a TPM is out of tolerance. The tolerance band is generally wider earlier in the life cycle and gets tighter as the system development progresses. Measured values that fall outside an established tolerance band alert management that it may be necessary to take corrective action.

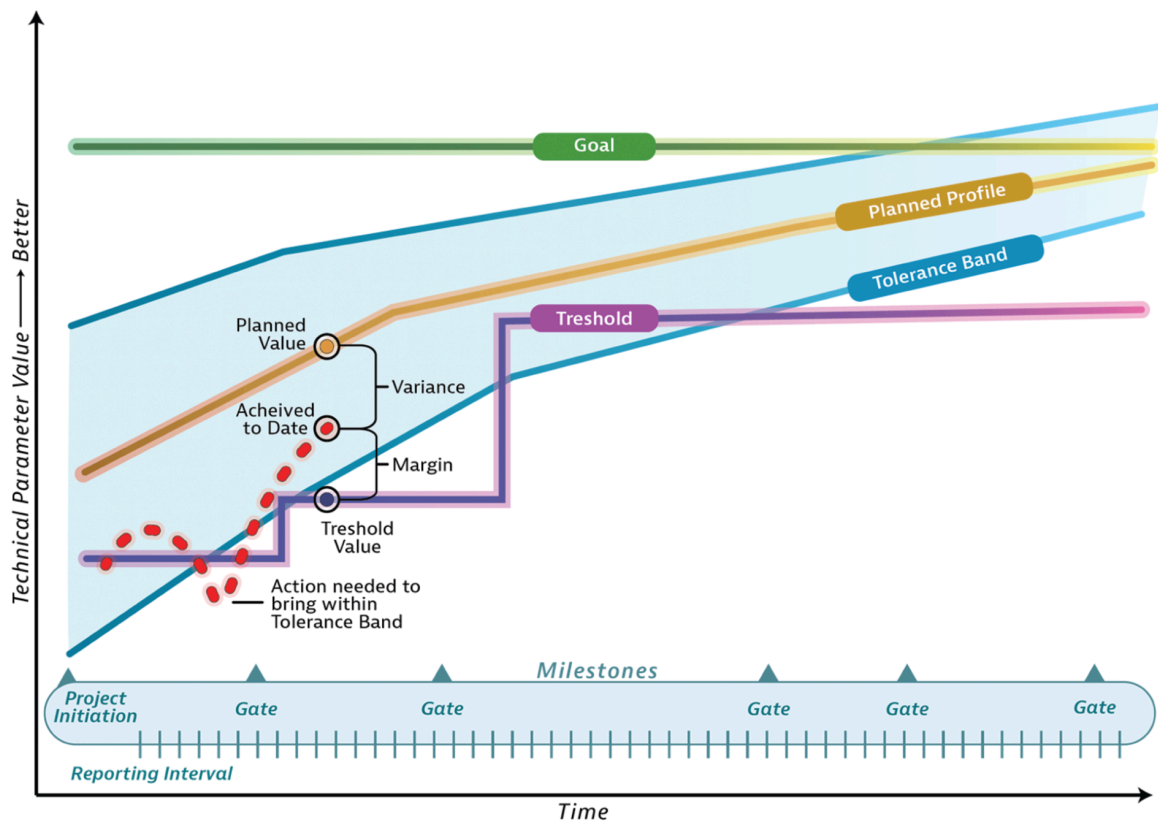


FIGURE 2.36 TPM monitoring. INCOSE SEH original figure created by Roedler and Walden. Usage per the INCOSE Notices page. All other rights reserved.

The progress of some TPMs relies on maturing a particular technology. Thus, it may be necessary to have a technology plan and technology readiness level (TRL) assessment as part of the input associated with a TPM.

2.3.4.8 Quality Assurance Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.3.8.1] The purpose of the Quality Assurance process is to help ensure the effective application of the organization's Quality Management process to the project.

Description Quality Assurance (QA) is broadly defined as the set of activities throughout the entire project life cycle necessary to provide adequate confidence that a product or service conforms to stakeholder requirements or that a process adheres to established methodology (ASQ,

2007). SE practitioners adopt and use QA processes as key contributors to project and systems development success. QA is a key aspect in QM from the project level to the individual processes. It involves the resourcing and improved performance of processes with built-in quality that is designed to prevent defects from occurring in delivered products and services.

Inputs/Outputs Inputs and outputs for the QA process are listed in [Figure 2.37](#). Descriptions of each input and output are provided in Appendix E.

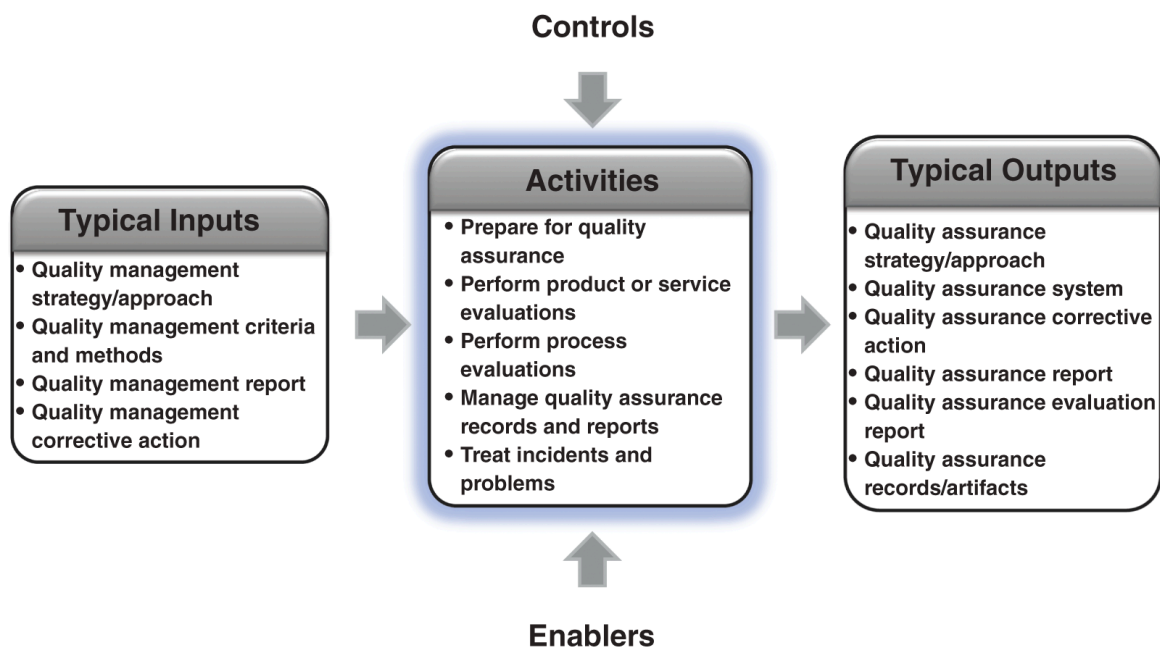


FIGURE 2.37 IPO diagram for the Quality Assurance process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Quality Assurance process includes the following activities:

- *Prepare for quality assurance.*
 - – Establish and maintain the QA strategy (often captured in a QA plan).
 - – Establish and maintain QA guidelines, policies, standards, and procedures.
 - – Define responsibilities and authorities.
- *Perform product or service evaluations.*
 - – Perform the evaluations at appropriate times in the life cycle as defined by the QA plan, ensuring V&V of the outputs of the life cycle

processes. Ensure that QA perspectives are appropriately represented during design, development, verification, validation, and production activities.

- – Evaluate product verification results as evidence of QA effectiveness.
- *Perform process evaluations.*
 - – Implement prescribed surveillance on processes to provide an independent evaluation of whether the developing organization is in compliance with established procedures.
 - – Evaluate enabling tools and environments for conformance and effectiveness.
 - – Flow applicable procedural and surveillance requirements throughout the project supply chain and evaluate subcontractor processes for conformance to allocated requirements.
- *Manage QA records and reports.*
 - – Create, maintain, and store records and reports in accordance with applicable requirements.
 - – Identify incidents and problems associated with product and process evaluations.
- *Treat incidents and problems.*

Note: Incidents are short-term anomalies or observations that require immediate attention, and problems are confirmed nonconformities that would cause the project to fail to meet requirements.

- – Document, classify, report, and analyze all anomalies.
- – Perform root cause analysis and note trends.
- – Recommend appropriate actions to resolve anomalies and errors, when indicated.
- – Track all incidents and problems to closure.

Common Approaches and Tips

- Management's commitment to QA is reflected in the integration of QM principles in the strategic planning and budgeting of the organization,

and the allocation of educational resources to achieve and sustain a reliable QM culture (see [Section 2.3.3.5](#)).

- A quality policy, mission, strategies, goals, and objectives provide essential inputs along with a description of an organization's fundamental values for quality assurance and the support of a growing QM culture.

Elaboration

QA Generally accepted theory and practice. QA is one of the two Quality Management (QM) defect prevention methods. The second is Quality Control which is described and contrasted in [Section 2.3.3.5](#). QA can be described as “putting good things into our processes” so that they perform as designed and conform to our stakeholder's requirements. Like QM, QA was born in the aerospace industry and was originally referred to as “reliability engineering.” It is generally associated with activities such as failure testing and pre-inspecting batches of materials and system elements that are then certified for use, thus preventing errors and defects from occurring by building-in quality. QA also includes infusing processes with reliable human resources and the appropriate policies, procedures, and training. W. Edwards Deming noted that “Quality comes not from inspection, but from improvement of the production process” (Deming, 1986).

QA Culture. “Ultimately, it is the people in an organization who can create a work culture in which quality is promoted and value is delivered to stakeholders” (Kennedy, 2005). An effective QA methodology defines competent, well-prepared humans as the major asset within processes that are then supported by the appropriate corporate environment, resources, and technologies to improve outcomes. It supports a high-performing work culture that diligently defines and fulfills stakeholder requirements with a Zero Defects Attitude (ZDA) (see [Section 2.3.3.5](#)) and is focused on continuous improvement. Philip Crosby noted that “Quality is the result of a carefully crafted cultural environment. It has to be the fabric of the organization, not part of the fabric” (Crosby, 1979).

The fabric of a QA-strengthened work culture is defined by fundamental skills and supporting values that create a sense of ownership by all participants. Workers who identify with an organization's core values have a stronger sense of psychological ownership and higher job satisfaction. At its core, psychological ownership is about an employee's possession and stewardship of an organization's core values and the pride they have about their enterprise/mission (Journal of Organizational Behavior, 2004). The workforce must have skills and experience that are directly related to the output objectives, and when skills are supported by shared values it creates a reliable work culture. (See [Section 2.3.3.5](#).) This strengthening of the work culture leads to greater employee engagement and naturally results in products and services with higher quality, along with other benefits to both the workforce and the corporation (Gallup, 2017, 2020). QM is an educational technology with systems, methods and language that help us reach our business goals and QA performs an essential resourcing, educational and process improvement role in ensuring that all elements of an organization execute its activities in accordance with its plans, and procedures as a means of building quality into products or services. While QA is focused on improving processes to prevent errors from occurring, QC provides an essential feedback loop to QA by providing defect rates and identifying their source in processes. By applying Work Process Analysis (WPA) to the defect data, QA can define and initiate input and process improvements to produce lean outcomes.

As the complexity of a project increases, the challenges to effectiveness and risk management also increase. These factors further emphasize the need for a coordinated QM culture with the proper balance of QC and QA along with the skills, experience, and values that align with the requirements of the project. Kennedy calls this properly configured alignment "Vocational Certainty," and that a high-performing work culture is measured by identifiable professional and personal attributes or values within an organization's workforce (Kennedy, 2005). Professional values for an effective QA educational initiative must build upon personal vocational certainty, and on administrative consistency that extends our attention to process details beyond the initial documentation of requirements and

progress reports. We must continue to interact with and challenge the stakeholders to mature their requirements so that stakeholder satisfaction can be assured.

2.3.5 Technical Processes

The ISO/IEC/IEEE 15288 includes 14 Technical Processes that are invoked concurrently, iteratively, and recursively throughout the system life cycle in conjunction with supporting agreement and technical management process activities. The Technical Processes are defined in ISO/IEC/IEEE 15288 as follows:

[5.7.5] The Technical Processes are used to define the requirements for a system, to transform the requirements into an effective product, to permit consistent reproduction of the product where necessary, to use the product, to provide the required services, to sustain the provision of those services and to dispose of the product when it is retired from service.

Technical Processes enable SE practitioners to coordinate the interactions between engineering specialists, other engineering disciplines, acquirers, operators, manufacturing/production and other system stakeholders. They also address conformance with the expectations and legislated requirements of society. These processes lead to the creation of a necessary and sufficient set of needs and requirements as well as resulting system solutions that address the needed capabilities within the bounds of performance, environment, external interfaces, ethical norms, societal expectations, regulations, and design constraints. Without the Technical Processes, the risk of project failure would be unacceptably high. [Figure 2.38](#) provides a graphical representation of the Technical Processes in context.

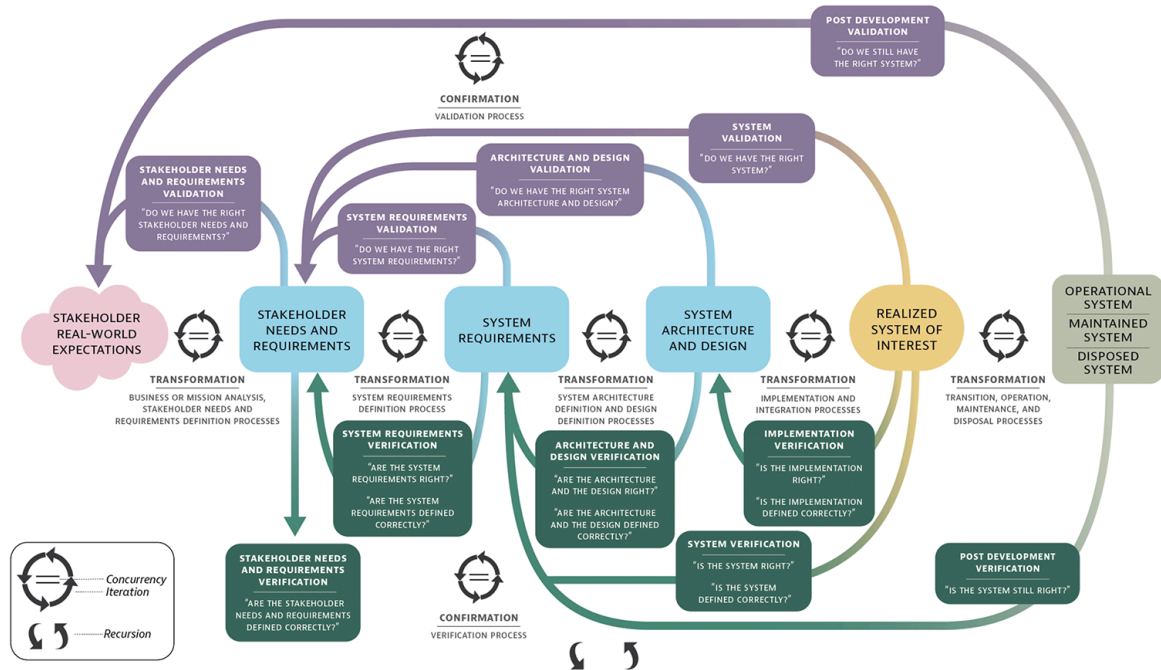


FIGURE 2.38 Technical Processes in context. INCOSE SEH original figure created by Roedler, Walden, and Wheatcraft derived from INCOSE NRM (2022). Usage per the INCOSE Notices page. All other rights reserved.

As shown in [Figure 2.38](#), at the beginning of the system life cycle are stakeholder real-world expectations for a SoI. The SoI could be the integrated system, a set of system elements, or a system element within the system architecture.

For each SoI, through a series of transformational actions across the life cycle, the technical processes transform input artifacts into output artifacts that are inputs into other technical processes, which in-turn transform those artifacts into additional artifacts. This series of transformations results in an SoI that addresses the capabilities needed by the stakeholders.

It is important to understand several key points for [Figure 2.38](#).

1. While the figure depicts the series of transformations in a linear fashion, in practice the Technical Processes are intended to be practiced concurrently, iteratively, and recursively as the project team moves down the layers of the system architecture. As such, the figure applies to each system element within the system architecture.
2. The Integration process is applied from the beginning of the project, managing the integrated system as the project team traverses the system

architecture. In doing so, the project team is continuously addressing interactions of the parts that make up the integrated system as well as interactions with the macro system of which it is a part. In addition, the project team is assessing the behavior of the system as a function of these interactions and looking for emerging properties—both good and bad—which is a key activity involved in Interface Management (see [Section 3.2.4](#)).

3. Following each transformation, the output artifacts are verified against the system requirements via the Verification process to ensure the output artifacts' transformation was “right” as defined by their requirements.
4. Following each transformation, the output artifacts are validated against the stakeholder needs and requirements via the Validation process to ensure the output artifacts are the “right” artifacts as defined by the stakeholder needs and requirements.
5. Once the SoI has been deployed and has entered into operations by its intended users, post-deployment validation is performed to help ensure the SoI remains the right SoI that meets the stakeholder real-world expectations—is the SoI still the right system? In addition, post-deployment verification is performed to help ensure the SoI is still meeting its requirements over time—is it still “right”?

Further elaboration of these key points is included within the following Technical Process sections.

New requirements can be placed on the SoI by the SoS configurations in which the SoI will participate. The SoS technical considerations apply to all the system life cycle processes across the life cycle stages, especially the Technical Processes that provide the concept and system definition (see [Section 4.3.6](#)).

2.3.5.1 Business or Mission Analysis Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.1.1] The purpose of the Business or Mission Analysis process is to define the overall strategic problem or opportunity, characterize the solution space, and determine potential solution class(es) that can address a problem or take advantage of an opportunity.

Description The Business or Mission Analysis process initiates the life cycle of the SoI by defining the problem or opportunity space; defining the mission, business, or operational problems or opportunities; identifying major stakeholders; characterizing the solution space by identifying environmental conditions and business constraints that bound the solution domain; identifying and prioritizing business needs; identifying and prioritizing business requirements, defining critical business success measures; developing preliminary life cycle concepts from the organizational perspectives including operations, acquisition, deployment, support, and retirement; and evaluating alternative solution classes and selecting a preferred solution class.

Inputs/Outputs Inputs and outputs for the Business or Mission Analysis process are listed in [Figure 2.39](#). Descriptions of each input and output are provided in Appendix E. Note that, as with all processes, the Business or Mission Analysis process is applied concurrently and iteratively evolving throughout the life cycle so that all SE artifacts mature as a result of the iterative application of the processes.

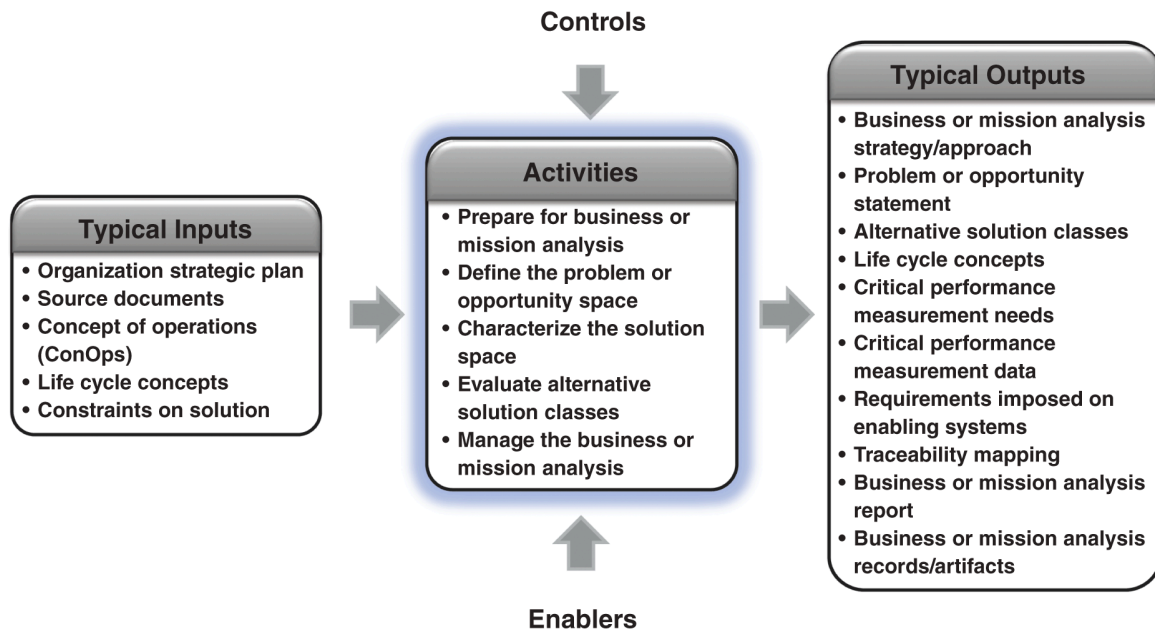


FIGURE 2.39 IPO diagram for Business or Mission Analysis process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Business or Mission Analysis process includes the following activities:

- *Prepare for business or mission analysis.*
 - Identify potential problems and opportunities resulting from changes in the organization’s strategy and Concept of Operations, while considering desired organization mission(s), goals, objectives, and other organizational business needs and business requirements. This may involve the development of concepts for a new solution but may also involve identifying gaps or deficiencies in existing capabilities, systems, products, or services and concepts for addressing those gaps or deficiencies.
 - Establish the strategy/approach for business or mission analysis. This involves the organizational approach(es) to defining the problem space, the characterization of the solution space, and the identification of an appropriate alternative solution classes.
 - Plan for the necessary enabling systems or services needed through the life cycle for business or mission analysis. This includes interfaces to organizational enabling systems or services such as business, acquisition, operations, production, project management, and SE tools and applications, financial systems, information

technology, databases, security controls, and other data and information repositories.

- – Ensure enabling system or service access needed to support business or mission analysis.
- *Define the problem or opportunity space.*
 - – Considering relevant trade-space factors as part of the problem and opportunity analysis. This analysis is focused on understanding the scope, drivers, constraints, risks, needs, and requirements associated with the problem or opportunity. The analysis includes changes in business needs, business requirements, opportunities, capabilities, performance improvement, security risks, safety risks, factors such as cost and effectiveness, value, regulation changes, standard changes, user dissatisfaction, lack of existing systems, and PESTEL factors (Political, Economic, Social, Technological, Environmental, and Legal). This may be accomplished through external, internal, or SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis.
 - – Define the problem or opportunity, mission, goals, objectives, and associated business needs and requirements to be addressed by a solution class. This definition is solution-class independent, since the solution could be an operational change, a change to an existing system or service, or a new system.
 - – Prioritize the problem or opportunity, mission, goals, objectives, needs, and requirements against other business needs and business requirements.
 - – Define critical business success measures. The business must define how it will know that the solution provided will meet its needs. Validation criteria establish critical and desired system performance—thresholds and objectives for system performance parameters that are critical for system success and those that are desired but may be subject to compromise to meet the critical parameters.
 - – Obtain agreement on the problem or opportunity, mission, goals, objectives, business needs, business requirements, and success measures.
- *Characterize the solution space.*

- – Define preliminary life cycle concepts for acquisition, development, deployment, operations, support, and retirement of the solution. Business stakeholders identify the stakeholders (individuals or groups) who are to be involved in any of the life cycle concepts. The life cycle concepts define what the system needs to do and how well from the business stakeholder's perspective of the intended use in the intended operational environment, when operated by the intended users in the context of all its life cycle activities, the required interactions with external systems, drivers and constraints, security, risks, business needs, and business requirements at the strategic level.
- – Establish a set of alternative classes spanning the potential solution space.
- *Evaluate alternative solution classes that span the potential solution space.*
 - – Evaluate the set of alternative solution concepts and select the preferred solution concepts against the organization's business needs, business requirements, and critical business success measures. Appropriate modeling, simulation, and analytical analysis will help determine the feasibility, value, and appropriateness of the alternative solution classes.
 - – Select the preferred solution class(es) and ensure each has been validated in the context of the proposed strategic level life cycle concepts. Feedback on feasibility, value, market factors, and alternatives is also provided for use in completing the definition of the organization's level life cycle concepts.
 - – Provide feedback to organization level life cycle concepts in terms of the selected solution class(es).
 - – Obtain agreement on the problem or opportunity statement, mission, goals, objectives, critical business success measures, life cycle concepts, business needs, and business requirements.
- *Manage the Business or Mission Analysis.*
 - – As key decisions are made, record the decision along with supporting information and rationale.

- – Establish and sustain traceability (analysis, rationale, and alternative solution classes).
- – Give CM the information items, work products, or other artifacts needed for baselines.

Common approaches and tips.

- Identify the enabling systems and materials needed for transition early in the life cycle to allow for the necessary lead time to obtain or access them.

Elaboration

Identify Major Stakeholders. Although the identification of stakeholders is undertaken at each stage of system development, during the Business or Mission Analysis process, business managers are responsible for nominating key stakeholders and are often responsible for establishing a stakeholder register and means of exchanging information. It is fundamentally a business management function to ensure stakeholders are available and able to contribute to the system development activities for the SoI—stakeholders are often occupied in other business operations activities and must be authorized in terms of both budget and time to expend the needed effort and resources on other than their current operational tasks.

Identify Business Needs and Requirements. For each problem or opportunity, it is important to identify the business needs and business requirements associated with needed capabilities, functionality, performance, and security as well as risk and compliance with standards and regulations. Business needs exist at several levels of abstraction, consist of identification of “what is needed” by the business to address the problem or opportunity, and can be communicated in several forms, such as the mission statement, goals, objectives, critical success measures, use cases, user stories, and individual need statements. The business requirements communicate what the business requires of the solution to address their needs without stating

a specific solution. The life cycle concepts are developed in response to the business needs and business requirements. Together, the business needs, business requirements, and critical business success measures communicate what is “necessary for acceptance” at the business level.

Life Cycle Concepts. Life cycle concepts address not only the concepts for the SoI during operations by the intended users in the operational environment, but also includes the concepts required to address the business needs, business requirements, critical business success measures, and higher-level stakeholder needs and stakeholder-owned system requirements across the system life cycle. Preliminary life cycle concepts are established and assured through the Business or Mission Analysis process to the extent needed to define the problem or opportunity space and characterize the solution space. Principal life cycle concepts include:

Concept of operations (ConOps)—Describes the way the organization will operate to achieve its missions, goals, and objectives. The ConOps captures how the system will potentially impact the acquiring and other organizations. “The ConOps describes the organization’s assumptions or intent in regard to an overall operation or series of operations of the business with using the system to be developed, existing systems and possible future systems. The ConOps serves as a basis for the organization to direct the overall characteristics of the future business and systems, for the project to understand its background, and for [its] users ... to implement the stakeholder requirements elicitation” (ISO/IEC/IEEE 29148, 2018) Ideally, the enterprise level ConOps should be an input to the Business or Mission Analysis process, but if it does not exist, it may need to be jointly developed and maintained. The ConOps also describes the higher-level system in which the SoI must operate.

Operational concept (OpsCon)—Describes the way the system will be used during operations, for what purpose, in its operational environment by its intended users and does not enable unintended users to negatively impact the intended use of the system nor allow unintended users from using the system in unintended ways. Also addressed are the needed capabilities, functionality, performance, quality, safety, security, compliance with standards and regulations, interactions with external systems, and operational risks. An OpsCon provides a user-oriented perspective that describes system characteristics of the to-be-delivered system. The OpsCon is used to communicate overall quantitative and qualitative system characteristics to the acquirer, user, supplier and other organizational elements.

Acquisition concept—For solutions that will be procured from a supplier, the acquisition concept describes the way the system will be acquired including aspects such as stakeholder engagement, needs definition, requirements definition, design, production, verification, validation, and contract deliverables. The supplier enterprise(s) may

need to develop more detailed concepts for production, assembly, verification, validation, transport of system, and/or system elements. For solutions that will be provided internal to the organization, the acquisition concept will include a production concept that describes the way the system will be developed and produced including aspects such as stakeholder engagement, needs definition, requirements definition, design, production, integration, verification, and validation.

Deployment concept—Describes the way the system will be delivered, integrated into its operational environment, and introduced into operations, including deployment considerations when the system will be integrated with other systems that are in operation and/or replace any systems in operation.

Support concept—Describes the logistics, desired support infrastructure and staffing considerations for supporting the system after it is deployed. A support concept would address operating support, engineering support, maintenance support, supply support, training support, and post-deployment verification and validation.

Retirement concept—Describes the way the system will be removed from operation and retired, including the disposal of any hazardous materials used in or resulting from the process and any legal obligations—for example, regarding IP rights protection, any external financial/ownership interests, sustainability, environmental impacts, and security concerns.

These preliminary life cycle concepts are defined first at the organizational level, to the extent required at that level, for the identified solution classes that address the problem or opportunity. The preliminary life cycle concepts are then elaborated and refined through the Stakeholder Needs and Requirements Definition process ([Section 2.3.5.2](#)). Through iteration, the life cycle concepts are refined throughout the life cycle as required as a result of feedback obtained through the conduct of the rest of the Technical Processes.

Uncertainties and risk. There will be uncertainties (see [Section 1.4.1](#)) in the preliminary life cycle concepts. Uncertainties can be related to differing stakeholder perspectives, business factors, market, management, technical performance, schedule, development and production costs, operations and support costs, security, and sustainability. These uncertainties are a source of risk. Each of these uncertainties need to be addressed using the Risk Management process ([Section 2.3.4.4](#)) in conjunction with the rest of the Technical Processes, especially the Stakeholder Needs and Requirements Definition process ([Section 2.3.5.2](#)), the System Requirements Definition process ([Section 2.3.5.3](#)), and the System Architecture Definition process ([Section 2.3.5.4](#)).

2.3.5.2 Stakeholder Needs and Requirements Definition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

[6.4.2.1] The purpose of the Stakeholder Needs and Requirements Definition process is to define the stakeholder needs and requirements for a system that can provide the capabilities needed by users and other stakeholders in a defined environment.

Description Successful projects depend on meeting the stakeholder real-world expectations as communicated by the needs and requirements of the stakeholders throughout the system life cycle. A stakeholder is any entity (individual or organization) with a legitimate interest in the system. Stakeholders exist at each of the levels of an organization and system architecture. The focus of the Stakeholder Needs and Requirements Definition process is on elaboration of the preliminary the life cycle concepts, on the stakeholder needs transformed from those concepts, and on the stakeholder requirements transformed from those needs. The activities during the process are constrained and driven by the preliminary life cycle concepts, business needs, business requirements, and critical

business success measures developed during the Business or Mission Analysis process (see [Section 2.3.5.1](#)).

In addition to identifying the stakeholders, this process elicits the operational use cases, scenarios, and life cycle concepts from stakeholders, identifies drivers and constraints, determines interactions with the operational and enabling systems, determines interactions with users and operators, characterizes the operational environment, and assesses risks associated with the development of a new or changed capability or new opportunities addressed by a solution class. The life cycle concepts are analyzed, matured, and transformed into a set of stakeholder needs. These needs are analyzed and transformed into a set of stakeholder requirements for the SoI. These stakeholder requirements communicate what the stakeholders expect from the SoI that will result in their needs being met using their terminology.

The stakeholder requirements drive and constrain the solution space by addressing stakeholder expectations for the SoI, characterizing the operational environment, and identifying external interface boundaries between the SoI and external systems across which there is an interaction. Traceability between the life cycle concepts, stakeholder needs, and stakeholder requirements is established as part of this process.

Stakeholder requirements govern the SoI's development and are an essential factor in further defining or clarifying the scope of the development project and elaborating on what is “necessary for acceptance.” If an organization is acquiring the system, this process provides the basis for the technical description of the deliverables in an agreement—typically in the form of a set of system requirements for a SoI and defined interfaces at the SoI boundaries.

Inputs/Outputs Inputs and outputs for the Stakeholder Needs and Requirements Definition process are shown in [Figure 2.40](#). Descriptions of each input and output are provided in Appendix E.

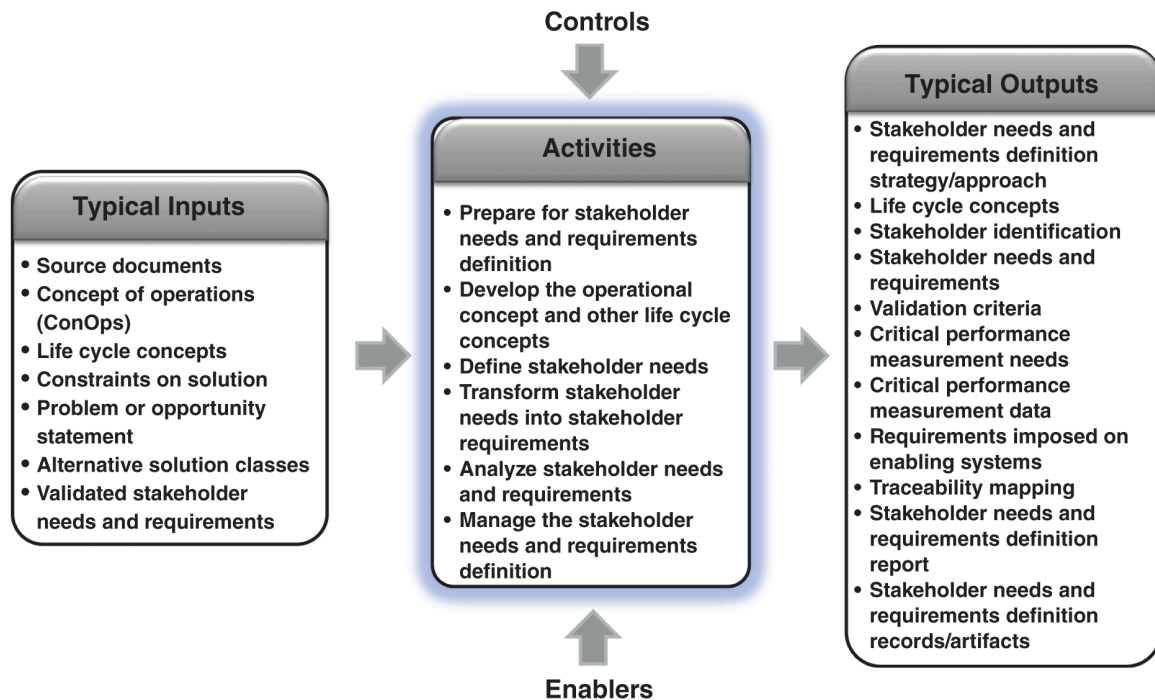


FIGURE 2.40 IPO diagram for Stakeholder Needs and Requirements Definition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Stakeholder Needs and Requirements Definition process includes the following activities:

- *Prepare for stakeholder needs and requirements definition.*
 - – Identify the stakeholders with an interest in the solution. Resolve differing interests. These stakeholders or classes of stakeholders will help identify constraints and define operational-level life cycle concepts, transform those concepts into operational-level stakeholder needs, which are then transformed into operational-level stakeholder requirements.
 - – Establish the strategy/approach for stakeholder needs and requirements definition. Understand the role and perspective of each stakeholder and identify any potential conflicts with other stakeholders to develop a stakeholder management plan and a strategy for defining life cycle concepts, stakeholder needs, and stakeholder requirements.
 - – Plan for the necessary enabling systems or services needed through the life cycle for stakeholder needs and requirements definition. Enabling systems may include tools for elicitation of stakeholder life

cycle concepts, recording drivers and constraints, defining risks, analysis, recording needs, recording requirements, and providing traceability between SE artifacts generated during this process, and managing those artifacts.

- – Ensure enabling system or service access needed to support stakeholder needs and requirement definition.
- *Develop the operational concept and other life cycle concepts.*
 - – Elaborate the operational concept (OpsCon) within the context of the concept of operations (ConOps), other life cycle concepts, and the preferred solution class(es). Preliminary life cycle concepts are developed in the Business or Mission Analysis process (see [Section 2.3.5.1](#))—in this process they are elaborated in more detail.
 - – Define a set of operational scenarios (or use cases) to identify required capabilities that correspond to anticipated operational and other life cycle concepts. Identify use cases and operational scenarios and associated capabilities, behaviors, and responses of the system or solution and environments across the SoI life cycle (concept, development, production, utilization, support, and retirement). The use cases and operational scenarios provide the information from the stakeholders needed to define the life cycle concepts; the range of intended uses of the system; the intended operational environment and the system's impact on the environment; intended users, and interfacing systems, platforms, or products.
 - – >Capture the characterization of the SoI's intended operational environment and users.
 - – Considering usability, identify user interactions. Capture factors (e.g., skills) that can affect the interactions.
 - – Identify external interface boundaries across which the SoI interacts.
 - – Considering the stakeholder and technical objectives and limitations, identify constraints on the solution.
 - – Identify risks associated with management, development, operations, and disposal, including misuse and loss scenarios (see Sections 3.1.12 and 3.1.13).
- *Define stakeholder needs.*

- – Identify stakeholder needs that reflect the intended life cycle concepts and associated constraints (see the INCOSE NRM [2022] and INCOSE GtNR [2022] for more details). Specify quality, health, safety, security, environment, compliance, and other stakeholder needs and functions that relate to critical qualities.
- – In conjunction with the Decision Management process, prioritize and select the essential needs. The System Analysis process (see [Section 2.3.5.6](#)) is used to analyze the life cycle concepts, resolve conflicts, and assess feasibility of those concepts. Needs are assessed and priorities are assigned in terms of criticality, value, completeness, correctness, consistency, security and feasibility.
- – As the stakeholder needs are selected, record the needs with their sources and rationale. Transform the life cycle concepts and other sources into a homogeneous, agreed-to integrated set of stakeholder needs. Establish traces from the needs to their sources. Although the stakeholder needs are not required to be as stated as rigorously as requirements, it is useful to follow the same rules as for quality requirements (see [Section 2.3.5.3](#) and the INCOSE GtWR [2023] for more details) since better-formed needs and sets of needs will result in less ambiguity in the transformation of the needs into requirements. For each stakeholder need, define attributes such as source, rationale, priority, and criticality.
- *Transform stakeholder needs into stakeholder requirements.*
 - – Transform the needs into stakeholder requirements. Define a set of stakeholder requirements consistent with the stakeholder needs.
 - – Identify any additional stakeholder requirements. Define stakeholder requirements that relate to safety, security, sustainability, human systems integration, etc. (see [Section 3.1](#)). Define stakeholder requirements that relate to high priority and critical functionality, performance, the operational environment, interactions with users, interactions with external interfacing and enabling systems, and compliance with standards and regulations. Ensure the stakeholder requirements are consistent with the life cycle concepts, needs, scenarios, interactions, constraints, operational risks, and SoI considerations.

- – Ensure high quality stakeholder requirements. Each stakeholder requirement should follow the rules for quality requirements and possesses characteristics such as necessary, singular, correct, unambiguous, feasible, appropriate to level, complete, conforming, and can be validated (see [Section 2.3.5.3](#) and the INCOSE GtWR (2023) for more details). For each stakeholder requirement, define attributes such as source, rationale, priority, and criticality.
- *Analyze stakeholder needs and requirements.*
 - – Analyze the complete sets of stakeholder needs and stakeholder requirements. Analyze the sets of stakeholder needs and stakeholder requirements to ensure they are correct, complete, consistent, comprehensible, appropriate to level, and feasible (see [Section 2.3.5.3](#) and the INCOSE GtWR [2023] for more details).
 - – Enable technical achievement monitoring through the definition of critical performance measures and quality characteristics.
 - – Define system validation criteria for each stakeholder need and requirement, the validation strategy, validation method, and responsible organization for providing evidence the stakeholder needs and requirements have been met.
 - – Review the analyzed stakeholder requirements with the applicable stakeholders to validate that their needs and expectations have been adequately captured and expressed.
 - – Resolve stakeholder needs and requirements issues. Negotiate changes, amendments, and modifications to resolve inconsistencies, conflicts, and unrealizable or impractical stakeholder needs and requirements.
- *Manage the stakeholder needs and requirements definition.*
 - – Obtain explicit agreement on the stakeholder needs and requirements.
 - – Establish and sustain traceability (stakeholder needs and requirements).
 - – Give CM the information items, work products, or other artifacts needed for baselines.
 - – Manage changes to the stakeholder needs and stakeholder requirements, as needed.